

Memory & Storage

RAM, ROM, Cache, SSD, USB Flash Drives, Cloud Storage

BSc Computer Science (First Year) · Detailed notes with diagrams · 2-mark & 14-mark questions

Learning objectives. After studying these notes you should be able to:

- Explain the memory hierarchy and the speed/size/cost trade-off.
- Distinguish RAM vs ROM and volatile vs non-volatile memory.
- Explain cache memory and describe a cache hit and a cache miss.
- Compare HDD, SSD, USB flash drives and cloud storage.
- Use the units of capacity from bit to terabyte.

1. Memory vs storage

Memory (primary) is fast, works directly with the CPU, and is mostly temporary. **Storage (secondary)** is slower, larger, cheaper, and permanent.

All the different kinds of memory in a computer sit on a ladder called the **memory hierarchy**. Moving up the ladder, memory gets **faster, smaller and costlier**; moving down, it gets **bigger, cheaper and slower**. No single technology can be fast, large and cheap all at once — the hierarchy is how computer designers get the best of all three.

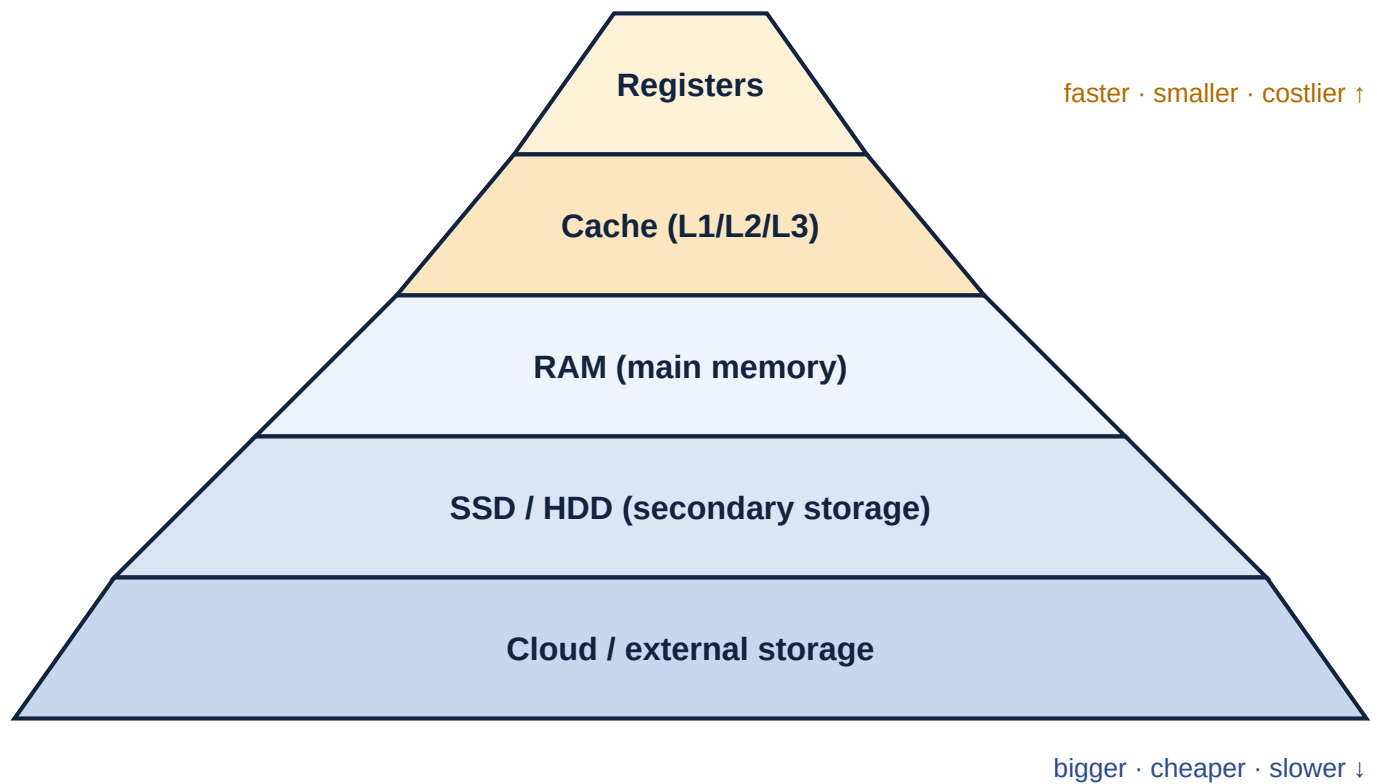


Fig 1. The memory hierarchy: registers and cache at top (fastest, smallest, dearest); cloud storage at the base (slowest, largest, cheapest).

Level	Approx. speed	Approx. capacity	Volatile?
Registers	~1 ns (fastest)	a few bytes	Yes
Cache (L1/L2/L3)	~1–10 ns	KB to tens of MB	Yes
RAM (main memory)	~50–100 ns	GBs	Yes
SSD / HDD	microseconds–milliseconds	hundreds of GB–TB	No
Cloud / external	network delay (slowest)	virtually unlimited	No

2. RAM — Random Access Memory

Definition. RAM is the computer's main working memory. When a program is opened, it is copied from storage into RAM so the CPU can access it quickly. It is called 'random access' because any memory location can be read or written in the same amount of time as any other.

- **Volatile** — it needs continuous power to hold its contents; switching off the computer erases everything in RAM.
- Holds the **operating system, running programs and their data** while the computer is on.
- More RAM lets a computer keep more programs open at once without slowing down.

- The common type used as main memory is **DRAM** (Dynamic RAM); a faster, costlier variant, **SRAM** (Static RAM), is used for cache.

3. ROM — Read Only Memory

Definition. ROM is **non-volatile** memory: it keeps its contents even without power. It is written once (usually at the time of manufacture) and is normally only *read* afterwards, not rewritten during ordinary use.

- Stores the computer's **permanent start-up instructions** — the BIOS/firmware and bootstrap program used to start the machine before the operating system loads.
- Variants: **PROM** (Programmable ROM, written once by the user), **EPROM** (Erasable Programmable ROM, erased using ultraviolet light), and **EEPROM** (Electrically Erasable Programmable ROM, which can be erased and rewritten electrically — the type used in modern firmware chips).

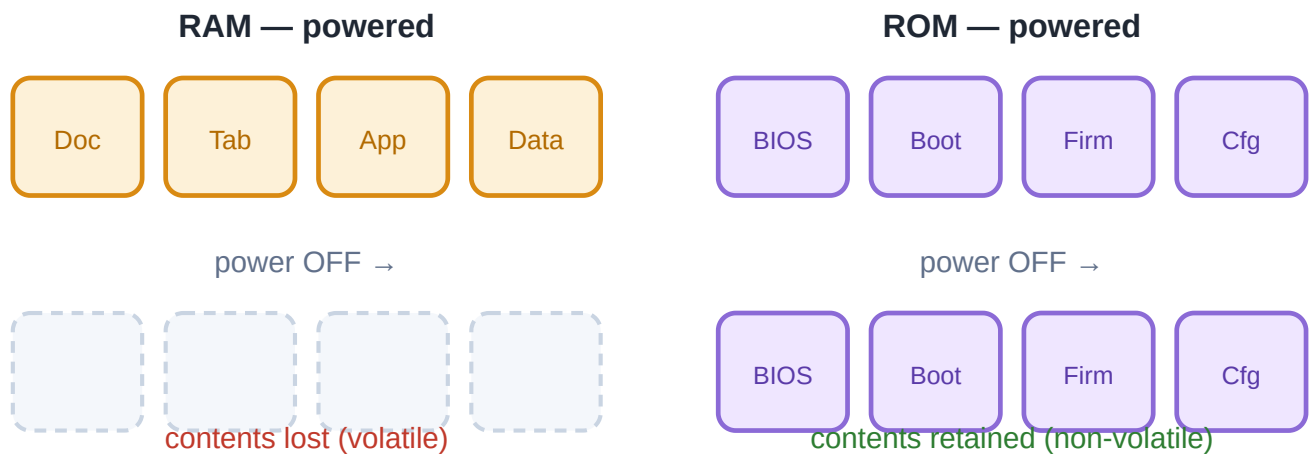


Fig 2. When power is cut: RAM's contents are lost (volatile); ROM's contents remain (non-volatile).

Feature	RAM	ROM
Full form	Random Access Memory	Read Only Memory
Volatility	Volatile (needs power)	Non-volatile (keeps data)
Operation	Read and write	Mainly read
Purpose	Temporary working memory for running programs	Permanent start-up instructions (BIOS/firmware)
Contents on power-off	Erased	Retained
Speed	Fast (read/write)	Fast to read

4. Cache memory

Definition. Cache is a small block of very fast memory placed between the CPU and RAM. It stores copies of the data and instructions the CPU is most likely to need again soon, so the CPU does not have to wait for the slower main memory every time.

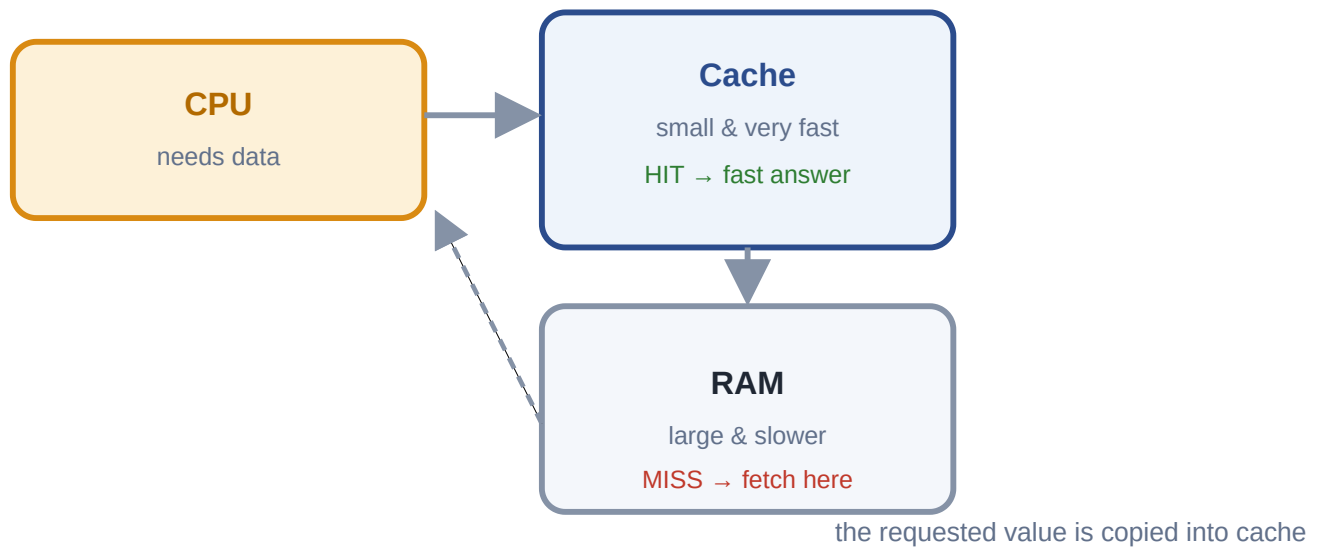
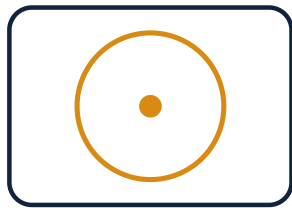


Fig 3. Cache sits between the CPU and RAM: hits are answered instantly; misses go to slower RAM and get copied into cache.

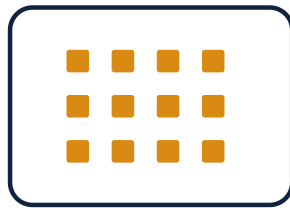
- **Cache hit** — the CPU finds the data it needs already in the cache; the request is served quickly.
- **Cache miss** — the data is not in the cache; the CPU must fetch it from the slower RAM, and a copy is then placed in the cache for next time.
- Cache works because of **locality of reference**: programs tend to reuse the same data, and data near something just used, repeatedly over a short time.
- Cache is organised in **levels**: **L1** is the smallest and fastest (closest to the CPU core), **L2** is larger and slightly slower, and **L3** is larger still and shared between CPU cores.

5. Secondary storage devices

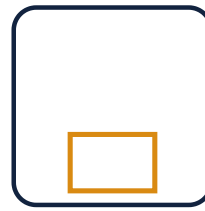
Secondary storage holds programs and files **permanently**, even when the power is off. It is not accessed directly by the CPU — data must first be loaded into RAM.



HDD



SSD



USB drive



Cloud

Fig 4. Four common secondary/external storage devices.

Device	How it stores data	Key traits
HDD (Hard Disk Drive)	Magnetically, on spinning platters read by a moving arm	Cheap, high capacity, but slower and has moving parts (more fragile).
SSD (Solid State Drive)	Electronically, in flash memory chips	No moving parts; much faster, quieter and more durable than an HDD; costs more per GB.
USB flash drive (pen drive)	Flash memory in a small, portable stick	Highly portable; convenient for moving files between computers.
Cloud storage	On remote servers (data centres) accessed via the internet	Accessible from anywhere; good for backup and sharing; speed depends on the internet connection.

6. Units of memory and storage

bit → byte → KB → MB → GB → TB → PB

each step is about 1024× the one before (8 bits = 1 byte)

Fig 5. The units of capacity, each about 1024 times the previous one.

Unit	Equivalent
1 bit	a single 0 or 1
1 byte	8 bits
1 Kilobyte (KB)	≈ 1,024 bytes
1 Megabyte (MB)	≈ 1,024 KB
1 Gigabyte (GB)	≈ 1,024 MB
1 Terabyte (TB)	≈ 1,024 GB
1 Petabyte (PB)	≈ 1,024 TB

Chapter summary. Memory and storage form a hierarchy that trades speed for size and cost. RAM (volatile) holds running programs; ROM (non-volatile) holds start-up firmware. Cache bridges the speed gap between the CPU and RAM using hits and misses. Secondary storage (HDD/SSD), portable storage (USB) and cloud storage keep data permanently, each with its own trade-off of speed, capacity, portability and cost.

2-Mark Questions (Short Answer)

Answer in 2–4 lines. These test definitions and single facts.

Q1. What is the memory hierarchy?**2 marks**

The memory hierarchy is the arrangement of different types of memory/storage in a computer from fastest and smallest (registers) to slowest and largest (cloud storage), trading speed against size and cost.

Q2. Define RAM.**2 marks**

RAM (Random Access Memory) is the computer's main working memory. It temporarily holds running programs and data, and can be read from or written to directly by the CPU.

Q3. Why is RAM called volatile?**2 marks**

RAM is called volatile because it requires continuous electrical power to retain data; as soon as power is removed, everything stored in it is lost.

Q4. Define ROM.**2 marks**

ROM (Read Only Memory) is non-volatile memory that permanently stores the computer's start-up instructions (BIOS/firmware). It keeps its data even without power and is mainly only read, not written.

Q5. Give one similarity and one difference between RAM and ROM.**2 marks**

Similarity: both are types of primary/main memory used by the CPU. Difference: RAM is volatile and read-write, while ROM is non-volatile and mainly read-only.

Q6. What is cache memory?**2 marks**

Cache memory is a small, very fast memory placed between the CPU and RAM that stores data the CPU is likely to need again soon, reducing the time spent waiting for slower RAM.

Q7. Differentiate between a cache hit and a cache miss.**2 marks**

A cache hit occurs when the requested data is found in the cache (fast). A cache miss occurs when the data is not in the cache and must be fetched from the slower RAM instead.

Q8. What is 'locality of reference'?**2 marks**

Locality of reference is the tendency of a program to access the same data, or data near it, repeatedly over a short period — this is why caching recently used data works well.

Q9. What is the main advantage of an SSD over an HDD?**2 marks**

An SSD has no moving parts and uses flash memory, making it significantly faster, quieter and more durable than an HDD, which stores data magnetically on spinning disks.

Q10. What is cloud storage?

2 marks

Cloud storage is a service where data is stored on remote servers (data centres) run by a provider and accessed over the internet, rather than on the user's own device.

Q11. Arrange in increasing order: MB, byte, GB, KB.

2 marks

byte < KB < MB < GB (each unit is about 1,024 times larger than the previous one).

Q12. What is a USB flash drive?

2 marks

A USB flash drive (pen drive) is a small, portable secondary storage device that uses flash memory and connects via a USB port, commonly used to transfer files between computers.

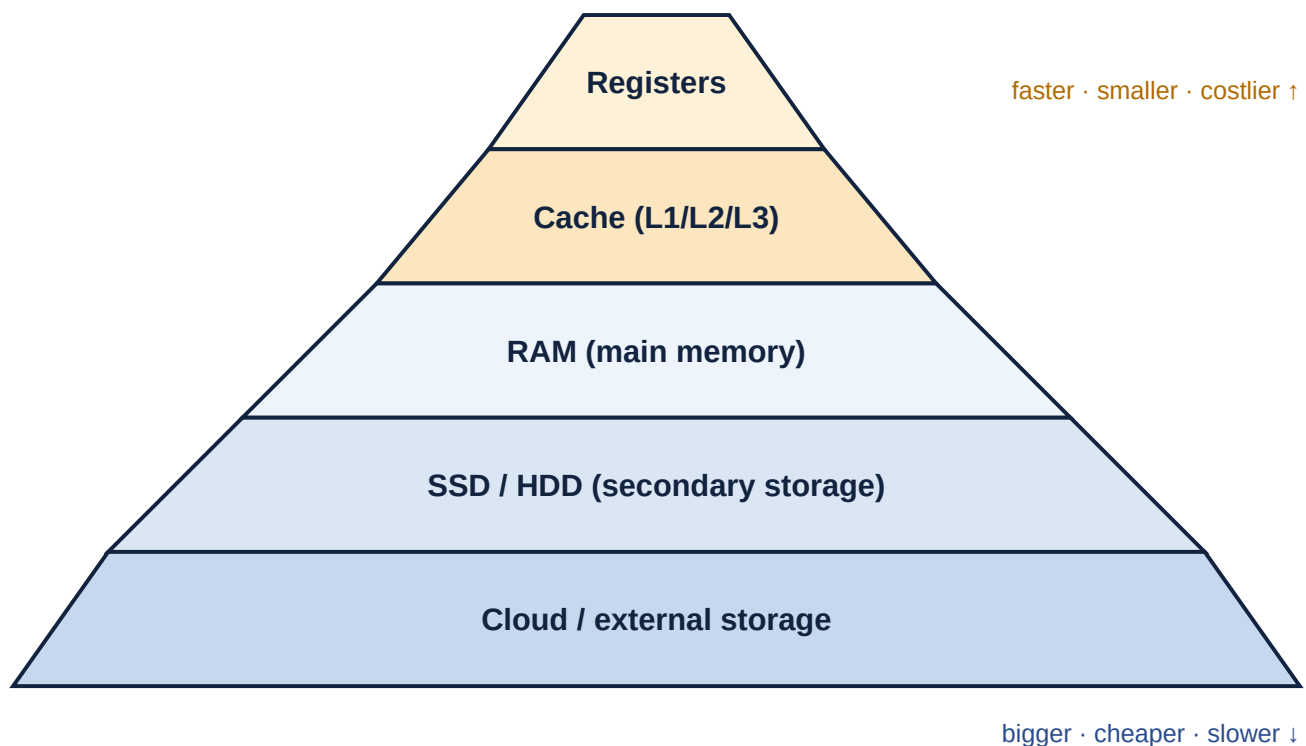
14-Mark Questions (Long Answer)

Long answers should include a definition, well-organised points, at least one labelled diagram, and a short conclusion.

Q1. Explain the memory hierarchy of a computer with a labelled diagram. Why is no single type of memory used for everything? 14 marks

Introduction. A computer needs memory that is fast, large and cheap — but no single technology can offer all three at once. Because of this, computer designers arrange several kinds of memory into a ladder called the **memory hierarchy**, giving the benefits of each where they matter most.

The hierarchy, top to bottom.



The memory hierarchy: fastest and smallest at top, slowest and largest at the base.

- **Registers** — inside the CPU itself; hold the values being processed this instant; extremely fast but tiny (a few bytes).
- **Cache (L1/L2/L3)** — very fast memory close to the CPU holding recently used data, bridging the speed gap to RAM.
- **RAM (main memory)** — holds the running programs and their data; fast, but volatile and moderate in size (GBs).
- **SSD/HDD (secondary storage)** — permanent, non-volatile storage for files and programs; much larger but slower.
- **Cloud/external storage** — remote storage accessed over a network; effectively unlimited but the slowest to reach.

Why not one technology for everything? Faster memory (like registers and cache) is built from more complex, expensive circuitry and can only be made in small quantities close to the CPU. Slower memory (like disks) uses cheaper technology that can be scaled to enormous sizes. By combining several levels, a computer gets the CPU's speed where it is needed most (registers/cache), reasonable working space in

RAM, and huge, affordable long-term storage in SSD/HDD and the cloud.

Conclusion. The memory hierarchy is a practical compromise: as we move down the pyramid we trade speed for capacity and lower cost, letting a computer be both fast and capable of storing enormous amounts of data.

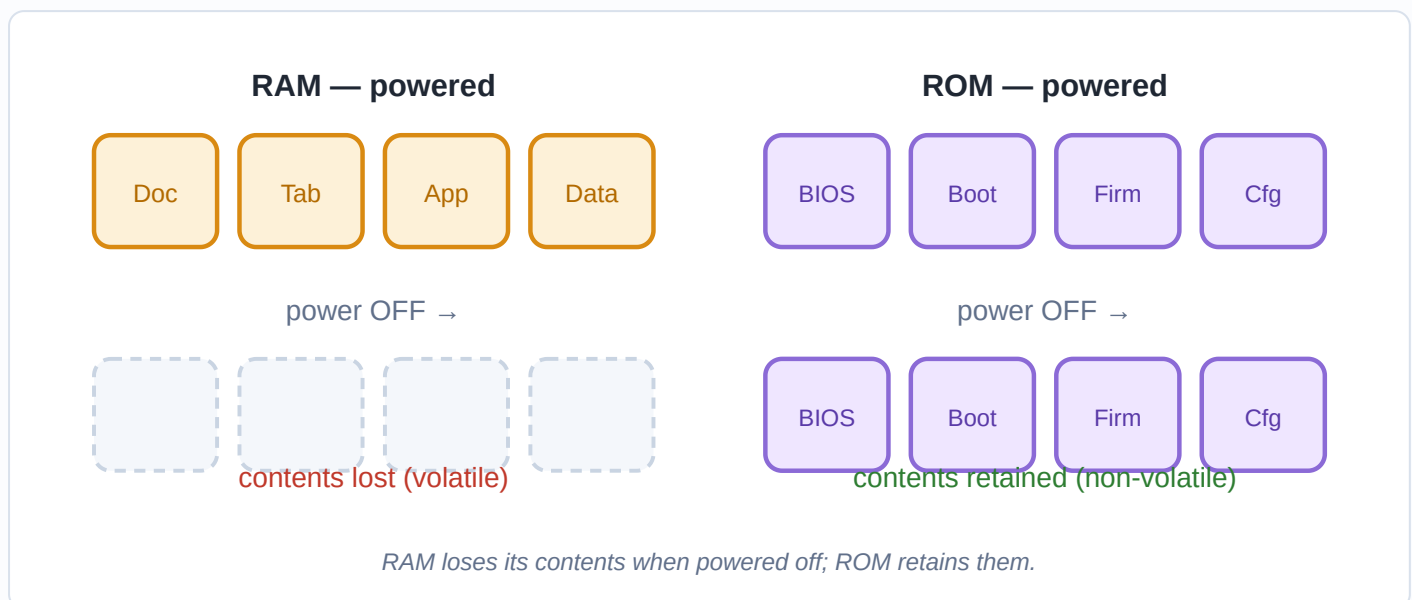
Q2. Differentiate between RAM and ROM. Explain their types and their role in a computer system.

14 marks

Introduction. RAM and ROM are both primary memory used directly by the CPU, but they serve very different purposes because of one key difference: volatility.

RAM (Random Access Memory). RAM is **volatile** memory that temporarily holds the operating system, running programs, and the data currently in use. The CPU can read from or write to any location in RAM in roughly the same time, which is why it is called 'random access'. When the computer is switched off, everything in RAM is lost. The common type used as main memory is **DRAM**; a faster but more expensive type, **SRAM**, is used in cache.

ROM (Read Only Memory). ROM is **non-volatile** memory that retains its contents even without power. It stores the computer's permanent start-up instructions — the BIOS/firmware needed to begin booting before the operating system loads. Variants include **PROM** (written once by the user), **EPROM** (erasable using ultraviolet light) and **EEPROM** (electrically erasable and rewritable, used in modern firmware chips).



Comparison table.

Feature	RAM	ROM
Volatility	Volatile	Non-volatile
Operation	Read and write	Mainly read
Purpose	Temporary working memory	Permanent start-up instructions
Data on power-off	Lost	Retained

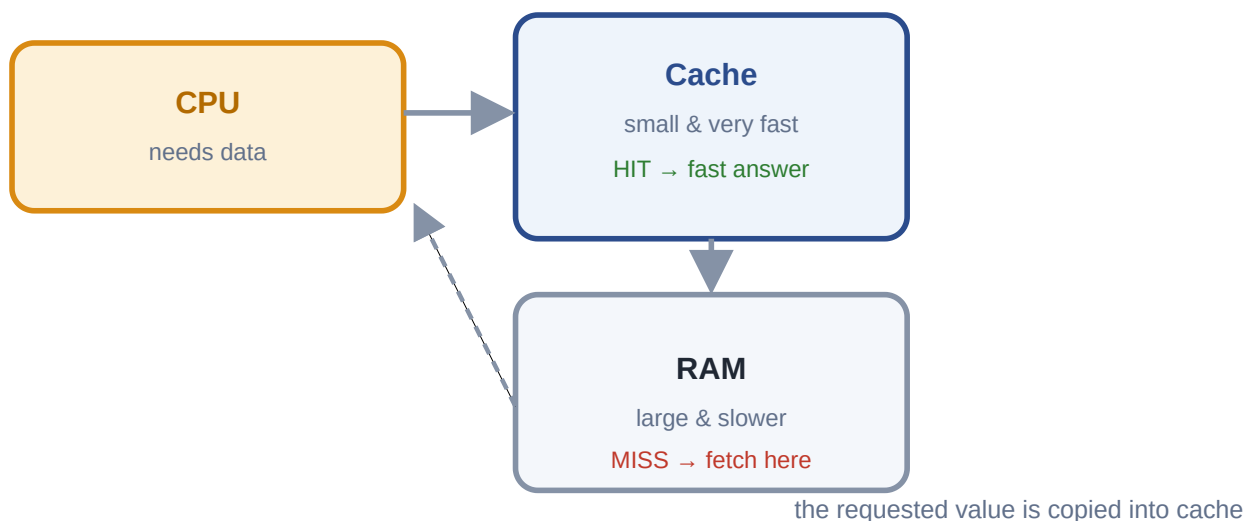
Conclusion. RAM gives the CPU fast, temporary working space for whatever is running now, while ROM guarantees the computer always has the permanent instructions it needs to start up — together they cover both the changing and the unchanging needs of the system.

Q3. What is cache memory? Explain how a cache hit and a cache miss work, and why cache memory improves overall system performance.

14 marks

Definition. Cache memory is a small amount of very fast memory placed between the CPU and main memory (RAM). Its purpose is to store copies of the data and instructions the CPU is likely to need again soon, so that the CPU does not have to wait for the much slower RAM every time.

How it works.



Cache sits between the CPU and RAM; hits are fast, misses go to RAM.

- **Cache hit.** When the CPU requests data, it first checks the cache. If the data is found there, this is a *hit*, and the CPU receives the data almost instantly.
- **Cache miss.** If the data is not present in the cache, this is a *miss*. The CPU must then fetch the data from the slower RAM. Once fetched, a copy of that data is placed into the cache, so that if it is needed again soon, it will now be a hit.

Why it works — locality of reference. Programs tend to reuse the same data and instructions repeatedly, and also tend to access data located near data they have just used. This pattern, called locality of reference, means that keeping recently used items in a small, fast cache results in most requests becoming hits over time.

Levels of cache. Modern CPUs use multiple levels: **L1** is the smallest and fastest, built into each CPU core; **L2** is larger and slightly slower; **L3** is larger still and often shared between all cores.

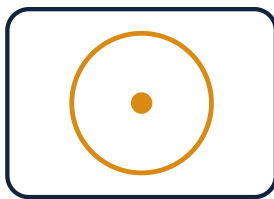
Effect on performance. Because most memory requests become fast cache hits rather than slow trips to RAM, the average time the CPU spends waiting for data drops sharply, letting the processor run closer to its full speed.

Conclusion. Cache memory is a small, fast buffer that exploits locality of reference to convert most memory accesses into quick hits, significantly improving the overall speed of the computer.

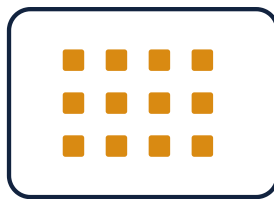
Q4. Compare HDD, SSD, USB flash drives and cloud storage as means of secondary storage, and discuss which is best suited to different situations.

14 marks

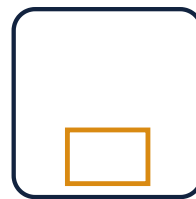
Introduction. Secondary storage keeps programs and files permanently, even when the computer is switched off. Several technologies exist, each with its own balance of speed, capacity, portability and cost.



HDD



SSD



USB drive



Cloud

Four common secondary and external storage devices.

The four types.

- **HDD (Hard Disk Drive).** Stores data magnetically on spinning platters read by a moving arm. It is cheap and offers very high capacity, but is slower than flash-based storage and has moving parts that make it more fragile.
- **SSD (Solid State Drive).** Stores data electronically in flash memory chips, with no moving parts. It is much faster, quieter and more durable than an HDD, though it costs more per gigabyte.
- **USB flash drive.** A small, portable device using flash memory, connected via a USB port. It is not meant for a computer's main storage but is extremely convenient for carrying and transferring files between machines.
- **Cloud storage.** Data is stored on remote servers belonging to a provider and accessed over the internet. It allows access from anywhere and is excellent for backup and sharing, but its speed depends entirely on the internet connection, and ongoing use is normally paid for as a subscription.

Comparison table.

Device	Speed	Capacity	Portability	Best suited for
HDD	Slower	Very large, cheap per GB	Low (internal)	Bulk, low-cost storage
SSD	Very fast	Large	Low–medium	Operating system & everyday use
USB drive	Moderate	Small–medium	Very high	Carrying/transferring files
Cloud	Depends on internet	Virtually unlimited	Access anywhere	Backup & sharing across devices

Conclusion. There is no single 'best' storage device — an SSD suits everyday speed, an HDD suits cheap bulk storage, a USB drive suits portability, and cloud storage suits backup and sharing; most modern

systems use a combination of these to meet all their needs.